

NCSA Control Catalogue

88 controls across 17 domains — generated 10 September 2026

1. What this document is

Every control the engine evaluates, with the framework identifiers each one maps to. It is generated from the rule definitions themselves, so it cannot state a control the engine does not run, or omit one that it does.

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2. Coverage at a glance

Controls	88
Domains	17
Mapped to NIST SP 800-53	88 of 88
Mapped to ISO/IEC 27001:2022	85 of 88
Carrying DISA STIG identifiers	16 of 88
With remediation commands	26 of 88
Conditional on another setting	9 of 88

By severity

Severity	Controls
Critical	6
High	30
Medium	39
Low	13

By tier

Tier	Controls	Applies to
Core	23	Applies to every managed network device, on every platform. A failure here is a finding on any estate.
Extended	49	Applies where the platform supports the capability. Resolves to NOT_APPLICABLE, with a stated reason, where it does not.
Category	16	Applies to a class of device -- firewall policy, cloud security group, layer-2 switching. Scoped by what the device is, not by vendor.

3. Standards this catalogue maps to

A control is written once, in vendor-neutral terms, and carries the identifiers of every framework that asks for it. One finding therefore answers several auditors at once.

Standard	Held	Role in this catalogue
NIST SP 800-53 Rev 5	1,196	The primary mapping. Public domain, so control text is stored in full. Every control here carries at least one identifier.
ISO/IEC 27001:2022	121	Annex A clause references, for organisations certifying against 27001. Clause number and short title only.

DISA STIG	553	Per-platform hardening identifiers. Attached by platform, because a STIG rule is specific to a device family.
CIS Benchmarks	3,506	Recommendation numbers, attached per platform in the vendor pack. Identifier only — the text is licensed.
NIST SP 800-171 Rev 3	130	Reached through the 800-53 crosswalk published in its own back-matter.
PCI DSS 4.0	279	Requirement identifiers, for cardholder-data environments.

Catalogue sizes are what this build has loaded, reported by the engine at `/frameworks`.

4. The controls

Grouped by domain, then ordered by severity. *Field* is the vendor-neutral name the control reads; *test* is the comparison applied to it.

authentication — 9 controls

Who may log in, and how they prove it

Control	Requirement	Field & test	Severity	Framework references
NCSA-AAA-001	Centralised AAA must be configured Centralised authentication gives per-administrator accountability and immediate revocation; local accounts give neither. Remediation held for: cisco_iosxe_router, juniper_srx	authentication.aaa_enabled equals True	High core	NIST 800-53: AC-2, IA-2 ISO 27001: A.5.16, A.5.18, A.8.2 STIG: V-215854, V-220565
NCSA-EXT-012	A privileged-mode secret must be set, not a reversible password 'enable password' is reversible; only 'enable secret' stores a hash.	authentication.enable_secret equals True	High extended	NIST 800-53: IA-5.1 ISO 27001: A.5.16, A.5.17
NCSA-EXT-013	Account lockout must be enabled Without lockout, credential guessing is limited only by network bandwidth. Remediation held for: sonicwall_sonicos	authentication.lockout.enabled equals True	High extended	NIST 800-53: AC-7 ISO 27001: A.8.5
NCSA-EXT-014	Administrative MFA should be required A single factor protecting a firewall's administrative interface is the weakest link in the whole control plane.	authentication.mfa.admin_required equals True	High extended	NIST 800-53: IA-2.1 ISO 27001: A.5.16
NCSA-EXT-015	An external authentication method list must be configured 'aaa new-model' alone proves nothing; a method list pointing at TACACS+ or RADIUS is what delivers per-administrator accountability.	authentication.aaa_authentication is set	High extended	NIST 800-53: IA-2, AC-2 ISO 27001: A.5.16, A.5.18, A.8.2
NCSA-PWD-001	Minimum password length must be enforced Short passwords are trivially brute-forced offline once a hash is obtained. 15 characters matches the current DISA baseline. Remediation held for: cisco_iosxe_router, sonicwall_sonicos	authentication.min_password_length gte 15	High core	NIST 800-53: IA-5, IA-5(1) ISO 27001: A.5.16, A.5.17 STIG: V-215826, V-220537
NCSA-PWD-003	Stored passwords must be encrypted Reversible password storage (e.g. Cisco type 7) is equivalent to storing the password in cleartext.	authentication.password_encryption equals True	High core	NIST 800-53: IA-5(1), SC-28 ISO 27001: A.5.10, A.5.16, A.5.17, A.5.33
NCSA-AAA-002	Only one local emergency account may exist A single break-glass account is required for AAA outage; additional local accounts bypass central control and audit. Remediation held for: cisco_iosxe_router	authentication.local_accounts max_count 1	Medium core	NIST 800-53: AC-2, AC-6 ISO 27001: A.5.15, A.5.16, A.5.18, A.8.18, A.8.2 STIG: V-215824, V-220535

NCSA-PWD-002	Password complexity must be enforced Complexity requirements reduce the effectiveness of dictionary attacks. Remediation held for: sonicwall_sonicos	authentication.complexity equals True	Medium core	NIST 800-53: IA-5(1) ISO 27001: A.5.16, A.5.17 STIG: V-215827, V-215828
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authorization — 2 controls

What an authenticated user may do

Control	Requirement	Field & test	Severity	Framework references
NCSA-AAA-003	Role-based authorisation must be in use Least privilege requires distinguishing operator from administrator.	authorization.role_based equals True	Medium core	NIST 800-53: AC-3, AC-6 ISO 27001: A.5.15, A.5.33, A.8.18, A.8.2, A.8.20, A.8.26, A.8.3, A.8.4 STIG: V-223186
NCSA-EXT-016	Command authorization must be configured Authentication says who; authorization says what they may run.	authorization.aaa_authorization is set	Medium extended	NIST 800-53: AC-3, AC-6 ISO 27001: A.5.15, A.5.33, A.8.18, A.8.2, A.8.20, A.8.26, A.8.3, A.8.4

banner — 2 controls

Legal notice presented before access

Control	Requirement	Field & test	Severity	Framework references
NCSA-BAN-001	A login banner must be displayed A banner establishes the legal basis for monitoring and for prosecuting unauthorised access. Remediation held for: cisco_iosxe_router, juniper_srx, sonicwall_sonicos	banner.login is set	Medium core	NIST 800-53: AC-8 ISO 27001: A.8.5 STIG: V-215814, V-220525, V-223189
NCSA-EXT-030	A MOTD banner should be present The MOTD banner carries the monitoring notice on pre-login connection.	banner.motd is set	Low extended	NIST 800-53: AC-8 ISO 27001: A.8.5

cloud — 2 controls

Cloud-native resource properties

Control	Requirement	Field & test	Severity	Framework references
NCSA-CAT-009	Egress traffic should be restricted Unrestricted egress is the exfiltration path in most cloud incidents.	cloud.security_groups.egress is set	Medium category	NIST 800-53: AC-4, SC-7 ISO 27001: A.5.14, A.8.16, A.8.20, A.8.22, A.8.23
NCSA-CAT-008	Cloud security groups must be identified Inventory precedes assessment.	cloud.security_groups.id is set	Low category	NIST 800-53: CM-8 ISO 27001: A.5.9, A.8.9

crypto — 4 controls

Cipher suites, key strength, protocol versions

Control	Requirement	Field & test	Severity	Framework references
NCSA-EXT-001	SSH must use strong key exchange and ciphers A strong protocol negotiated down to SHA-1 key exchange or a CBC cipher provides no meaningful protection.	crypto.weak_ciphers max_count 0	High extended	NIST 800-53: SC-8.1, SC-13 ISO 27001: A.5.10, A.5.14, A.5.33, A.8.20, A.8.24, A.8.26

NCSA-EXT-005	RSA host key must be at least 2048 bits Keys below 2048 bits are no longer considered computationally infeasible to break.	crypto.rsa_modulus gte 2048	High extended	NIST 800-53: SC-13 ISO 27001: A.8.24, A.8.26
NCSA-HTTPS-001	HTTPS management must use TLS 1.2 or higher TLS below 1.2 has known weaknesses and is no longer considered adequate for administrative sessions. Conditional: HTTPS management is disabled, so its TLS version is moot	crypto.tls.minimum_version in 1.2, 1.3	High core	NIST 800-53: SC-8(1), SC-13 ISO 27001: A.5.10, A.5.14, A.5.33, A.8.20, A.8.24, A.8.26
NCSA-SSH-003	Weak SSH ciphers must not be permitted A strong protocol negotiated down to a weak cipher provides no protection.	crypto.weak_ciphers max_count 0	Medium core	NIST 800-53: SC-8(1), SC-13 ISO 27001: A.5.10, A.5.14, A.5.33, A.8.20, A.8.24, A.8.26 STIG: V-215844, V-220555

exposure — 3 controls

What the policy exposes to an untrusted network

Control	Requirement	Field & test	Severity	Framework references
NCSA-CLD-001	Administrative ports must not be open to the internet An administrative port reachable from 0.0.0.0/0 is exposed to continuous automated scanning. This is the most frequently exploited cloud misconfiguration and needs no further vulnerability to be dangerous.	exposure.admin_ports_open_to_internet max_count 0	Critical category	NIST 800-53: AC-17(1), SC-7, SC-7(5) ISO 27001: A.5.14, A.6.7, A.8.16, A.8.20, A.8.22, A.8.23
NCSA-CLD-002	No rule may permit all protocols from the internet An any/any ingress rule defeats the purpose of the security group and silently exposes every service the instance ever runs.	exposure.any_any_rules max_count 0	Critical category	NIST 800-53: AC-4, SC-7, SC-7(5) ISO 27001: A.5.14, A.8.16, A.8.20, A.8.22, A.8.23
NCSA-CLD-003	Unrestricted ingress must be justified and minimal Public services legitimately need open ingress; the count is reported so each instance can be reviewed rather than assumed correct.	exposure.unrestricted_ingress max_count 0	Medium category	NIST 800-53: AC-4, SC-7 ISO 27001: A.5.14, A.8.16, A.8.20, A.8.22, A.8.23

firewall — 8 controls

Rule content, ordering and resolvability

Control	Requirement	Field & test	Severity	Framework references
NCSA-CAT-002	Firewall rules must not permit any source to any destination An any-source rule defeats the purpose of the policy it sits in.	firewall.rules.source contains_none 0.0.0.0/0, any	Critical category	NIST 800-53: AC-4, SC-7.5 ISO 27001: A.5.14, A.8.16, A.8.20, A.8.22, A.8.23
NCSA-CAT-004	All policy references must resolve A dangling group or service reference means part of the policy could not be evaluated; the assessment is incomplete until it is fixed.	firewall.rules_unevaluable max_count 0	High category	NIST 800-53: CM-6, AC-4 ISO 27001: A.5.14, A.8.22, A.8.23, A.8.9
NCSA-CAT-007	Defined ACLs must actually be applied An ACL that exists but is bound to no interface enforces nothing -- a common and easily missed misconfiguration.	firewall.acl_applied is set	High category	NIST 800-53: AC-4 ISO 27001: A.5.14, A.8.22, A.8.23

NCSA-CLD-004	Default deny must be the implicit policy Anything not explicitly permitted must be denied.	firewall.default_action equals deny	High category	NIST 800-53: AC-4, SC-7(5) ISO 27001: A.5.14, A.8.16, A.8.20, A.8.22, A.8.23
NCSA-CAT-001	Firewall rules must be logged An unlogged permit rule leaves no evidence that traffic was allowed.	firewall.rules.log equals True	Medium category	NIST 800-53: AU-2, AC-4 ISO 27001: A.5.14, A.8.15, A.8.22, A.8.23
NCSA-CAT-003	Every rule must resolve to a concrete service A rule whose service cannot be resolved cannot be assessed for exposure.	firewall.rules.service is set	Medium category	NIST 800-53: AC-4 ISO 27001: A.5.14, A.8.22, A.8.23
NCSA-CAT-005	Security zones must be defined Zones are the trust boundary the whole policy is expressed against.	firewall.zones is set	Medium category	NIST 800-53: SC-7 ISO 27001: A.5.14, A.8.16, A.8.20, A.8.22, A.8.23
NCSA-CAT-006	Access control lists must be defined A device with no ACLs is forwarding without policy.	firewall.acls is set	Medium category	NIST 800-53: AC-4 ISO 27001: A.5.14, A.8.22, A.8.23

interfaces — 3 controls

Physical and logical interface state

Control	Requirement	Field & test	Severity	Framework references
NCSA-CAT-010	Interfaces must be assigned to a zone An interface outside any zone has undefined policy applied to it.	interfaces.zone is set	Medium category	NIST 800-53: SC-7 ISO 27001: A.5.14, A.8.16, A.8.20, A.8.22, A.8.23
NCSA-CAT-011	Unused interfaces should be shut down An enabled unused port is an unmonitored entry point.	interfaces.shutdown is set	Low category	NIST 800-53: CM-7 ISO 27001: A.8.19
NCSA-EXT-042	Interfaces should carry a description Undocumented interfaces are how forgotten paths survive audits.	interfaces.description is set	Low extended	NIST 800-53: CM-8 ISO 27001: A.5.9, A.8.9

l2 — 3 controls

Layer-2 protections

Control	Requirement	Field & test	Severity	Framework references
NCSA-EXT-034	DHCP snooping should be enabled on access switches Without snooping, any host can offer DHCP and become the default gateway.	l2.dhcp_snooping equals True	Medium extended	NIST 800-53: SC-7 ISO 27001: A.5.14, A.8.16, A.8.20, A.8.22, A.8.23
NCSA-EXT-035	Dynamic ARP inspection should be enabled DAI is what stops ARP poisoning turning a switched segment into a shared one.	l2.arp_inspection equals True	Medium extended	NIST 800-53: SC-7 ISO 27001: A.5.14, A.8.16, A.8.20, A.8.22, A.8.23
NCSA-EXT-036	IP source guard should be enabled Source guard prevents address spoofing from an access port.	l2.ip_source_guard equals True	Low extended	NIST 800-53: SC-7 ISO 27001: A.5.14, A.8.16, A.8.20, A.8.22, A.8.23

logging — 9 controls

What is recorded, and where it is sent

Control	Requirement	Field & test	Severity	Framework references
NCSA-EXT-022	Remote syslog must be configured Logs held only on the device are lost precisely when they matter. Remediation held for: sonicwall_sonicos	logging.remote_syslog equals True	High extended	NIST 800-53: AU-4, AU-9 ISO 27001: A.5.33, A.8.15, A.8.6

NCSA-LOG-001	Logging must be enabled Without local logging there is no record of administrative action. Remediation held for: cisco_iosxe_router, juniper_srx	logging.enabled equals True	High core	NIST 800-53: AU-2, AU-3, AU-12 ISO 27001: A.5.28, A.8.15
NCSA-EXT-017	Command accounting must be enabled Without command accounting there is no record of what an administrator did.	logging.command_accounting equals True	Medium extended	NIST 800-53: AU-2, AU-12 ISO 27001: A.8.15
NCSA-EXT-018	Logging severity must capture informational events A severity threshold set too high discards the events an investigation needs.	logging.severity is set	Medium extended	NIST 800-53: AU-2 ISO 27001: A.8.15
NCSA-EXT-020	Persistent logging should survive a reboot Buffer-only logging is erased by the reboot an attacker causes.	logging.persistent equals True	Medium extended	NIST 800-53: AU-9 ISO 27001: A.5.33, A.8.15
NCSA-EXT-021	Denied traffic must be logged Denied traffic is the signal; without it a blocked attack is invisible.	logging.log_denied equals True	Medium extended	NIST 800-53: AU-2, AC-4 ISO 27001: A.5.14, A.8.15, A.8.22, A.8.23
NCSA-LOG-002	At least two remote syslog servers must be configured Logs held only on the device are lost precisely when they matter -- on compromise. Two collectors survive one failing. Remediation held for: cisco_iosxe_router, juniper_srx, sonicwall_sonicos	logging.servers min_count 2	Medium core	NIST 800-53: AU-4, AU-9, AU-6 ISO 27001: A.5.25, A.5.33, A.6.8, A.8.15, A.8.6 STIG: V-220139, V-220568
NCSA-CAT-012	NetFlow/traffic export should be enabled Flow export is what makes retrospective investigation possible.	logging.netflow equals True	Low category	NIST 800-53: AU-12, SI-4 ISO 27001: A.8.15, A.8.16
NCSA-EXT-019	Logs must be sourced from a stable interface A fixed source address keeps log correlation stable across path changes.	logging.source_interface is set	Low extended	NIST 800-53: AU-3 ISO 27001: A.5.28, A.8.15

management — 17 controls

The administrative plane and its transports

Control	Requirement	Field & test	Severity	Framework references
NCSA-TEL-001	Telnet must be disabled Telnet transmits credentials and session content in cleartext on the management path. Remediation held for: cisco_iosxe_router, juniper_srx	management.telnet.enabled equals False	Critical core	NIST 800-53: AC-17(2), SC-8, SC-8(1) ISO 27001: A.5.10, A.5.14, A.5.33, A.6.7, A.8.20, A.8.26 STIG: V-223209, V-228655, V-228669
NCSA-VTY-001	Management lines must accept SSH only Disabling the telnet daemon is insufficient if a VTY line still accepts telnet transport. Remediation held for: cisco_iosxe_router	management.vty.transport_input contains_none telnet, all, lat, rlogin	Critical core	NIST 800-53: AC-17(2), SC-8(1) ISO 27001: A.5.10, A.5.14, A.5.33, A.6.7, A.8.20, A.8.26

NCSA-HTTP-001	Cleartext HTTP management must be disabled An unencrypted web management interface exposes administrative credentials to anyone on the path. Remediation held for: cisco_iosxe_router, juniper_srx	management.http.enabled equals False	High core	NIST 800-53: AC-17(2), SC-8, SC-8(1) ISO 27001: A.5.10, A.5.14, A.5.33, A.6.7, A.8.20, A.8.26 STIG: V-215823, V-220534
NCSA-SSH-001	SSH must be enabled for management access Encrypted management is the precondition for removing cleartext access; remove Telnet only after SSH is proven working. Remediation held for: cisco_iosxe_router, juniper_srx	management.ssh.enabled equals True	High core	NIST 800-53: AC-17, AC-17(2), SC-8 ISO 27001: A.5.10, A.5.14, A.5.33, A.6.7, A.8.20, A.8.26
NCSA-SSH-002	SSH must use protocol version 2 SSHv1 has known cryptographic weaknesses and cannot be considered confidential. Only version 2 is acceptable. Remediation held for: cisco_iosxe_router, juniper_srx	management.ssh.version equals 2	High core	NIST 800-53: AC-17(2), SC-8(1), IA-5 ISO 27001: A.5.10, A.5.14, A.5.16, A.5.17, A.5.33, A.6.7, A.8.20, A.8.26 STIG: V-215844, V-220555, V-223209
NCSA-TIME-001	Idle session timeout must be 10 minutes or less An unattended authenticated session is an unauthenticated path to administrative control. Remediation held for: cisco_iosxe_router, juniper_srx, sonicwall_sonicos	management.vty.exec_timeout lte 10	High core	NIST 800-53: AC-12, SC-10 ISO 27001: A.8.20 STIG: V-215833, V-220544
NCSA-VTY-002	Management access must be restricted by ACL Restricting the management plane to known source networks removes the device from opportunistic internet-wide scanning. Remediation held for: cisco_iosxe_router	management.vty.access_class is set	High core	NIST 800-53: AC-3, AC-4, SC-7 ISO 27001: A.5.14, A.5.15, A.5.33, A.8.16, A.8.18, A.8.20, A.8.22, A.8.23, A.8.26, A.8.3, A.8.4 STIG: V-215812, V-220523
NCSA-EXT-002	SSH strict host key checking should be enabled Without host key verification an administrator cannot detect an interposed device on the management path.	management.ssh.strict_host_key equals True	Medium extended	NIST 800-53: SC-23
NCSA-EXT-003	SSH must be bound to a management source interface Sourcing SSH from a dedicated management interface keeps the control plane off transit paths.	management.ssh.source_interface is set	Medium extended	NIST 800-53: SC-7 ISO 27001: A.5.14, A.8.16, A.8.20, A.8.22, A.8.23
NCSA-EXT-006	SSH idle sessions must time out An idle authenticated SSH session is an unattended path to the control plane.	management.ssh.timeout lte 120	Medium extended	NIST 800-53: AC-12, SC-10 ISO 27001: A.8.20
NCSA-EXT-007	HTTPS management must be enabled where web management is used If any web management is offered it must be the encrypted one.	management.https.enabled equals True	Medium extended	NIST 800-53: SC-8.1 ISO 27001: A.5.10, A.5.14, A.5.33, A.8.20, A.8.26
NCSA-EXT-008	HTTP management must redirect to HTTPS A redirect prevents an administrator from silently transiting cleartext.	management.http.redirect_https equals True	Medium extended	NIST 800-53: SC-8.1 ISO 27001: A.5.10, A.5.14, A.5.33, A.8.20, A.8.26

NCSA-EXT-009	The auxiliary port must not offer an EXEC session The aux port is a documented back door that survives most hardening reviews.	management.aux.exec_disabled equals True	Medium extended	NIST 800-53: AC-17 ISO 27001: A.5.14, A.6.7
NCSA-EXT-010	Management lines must not permit outbound sessions Outbound transports let a compromised device pivot from the control plane.	management.vty.transport_output contains_none telnet, all, lat, rlogin	Medium extended	NIST 800-53: AC-4 ISO 27001: A.5.14, A.8.22, A.8.23
NCSA-SSH-004	SSH authentication attempts must be limited Limiting retries per connection raises the cost of credential guessing. Remediation held for: cisco_iosxe_router, juniper_srx	management.ssh.max_auth_tries lte 3	Medium core	NIST 800-53: AC-7 ISO 27001: A.8.5
NCSA-EXT-004	Public-key authentication should be available for SSH Key-based administrator authentication removes the reusable shared secret.	management.ssh.pubkey_auth equals True	Low extended	NIST 800-53: IA-2.1 ISO 27001: A.5.16
NCSA-EXT-032	ICMP unreachable rate limiting should be enabled Rate limiting unreachables reduces the device's value as a reconnaissance oracle.	management.icmp_ratelimit equals True	Low extended	NIST 800-53: SC-5

platform — 11 controls

Appliance-level properties and the management API

Control	Requirement	Field & test	Severity	Framework references
NCSA-API-001	Management API must not accept HTTP basic authentication Basic authentication sends the administrator credential on every request, recoverable by anyone who can observe or replay the session. Conditional: the management REST API is disabled, so its authentication settings are moot	platform.api.basic_auth equals False	High extended	NIST 800-53: IA-2, IA-5, SC-8 ISO 27001: A.5.15, A.5.17, A.8.5
NCSA-API-002	Management API digest authentication must not use MD5 MD5 is collision-prone and unfit for authentication; SHA-256 digest is available on the same interface. Conditional: the management REST API is disabled, so its authentication settings are moot	platform.api.digest_md5 equals False	High extended	NIST 800-53: IA-7, SC-13 ISO 27001: A.8.24
NCSA-API-003	Management API public keys must be at least 2048 bits A 1024-bit RSA key is below every current recommendation and is considered breakable by a well-resourced attacker. Conditional: the management REST API is disabled, so its authentication settings are moot	platform.api.pubkey_bits gte 2048	High extended	NIST 800-53: IA-5(2), SC-12, SC-13 ISO 27001: A.8.24

NCSA-PLT-002	Stored credentials and configuration parameters must be encrypted Credentials held in the saved configuration are readable by anyone who obtains a backup. Remediation held for: sonicwall_sonicos	platform.stored_credentials_encrypted equals True	High extended	NIST 800-53: IA-5(1), SC-28 ISO 27001: A.5.17, A.8.24
NCSA-API-004	Management API must not permit unrestricted cross-origin requests Cross-origin access lets a page the administrator merely visits drive the firewall using their session. Conditional: the management REST API is disabled, so its authentication settings are moot	platform.api.cors equals False	Medium extended	NIST 800-53: AC-4, SC-7 ISO 27001: A.8.20, A.8.23
NCSA-API-005	Management API session security must be enabled Without session binding an intercepted API session can be resumed from elsewhere. Conditional: the management REST API is disabled, so its authentication settings are moot	platform.api.session_security equals True	Medium extended	NIST 800-53: AC-12, SC-23 ISO 27001: A.8.5
NCSA-API-006	Management API access must be logged Configuration changes made through the API are invisible to review if the API does not log. Conditional: the management REST API is disabled, so its authentication settings are moot	platform.api.logging equals True	Medium extended	NIST 800-53: AU-2, AU-12 ISO 27001: A.8.15
NCSA-EXT-011	The REST/management API must be deliberately controlled An enabled management API is remote administrative surface and must be a conscious decision, not a default.	platform.api.enabled is set	Medium extended	NIST 800-53: AC-17, CM-7 ISO 27001: A.5.14, A.6.7, A.8.19
NCSA-EXT-029	Configuration change archiving should be enabled Without archiving there is no record of what changed or the ability to roll back. Remediation held for: sonicwall_sonicos	platform.config_archive equals True	Medium extended	NIST 800-53: CM-3, CM-6 ISO 27001: 8.1, 9.3.3, A.8.32, A.8.9
NCSA-EXT-041	Firmware version must be recorded for vulnerability tracking Without a recorded firmware version, no CVE can be matched to this device.	platform.firmware.version is set	Medium extended	NIST 800-53: CM-8, RA-5 ISO 27001: A.5.9, A.8.8, A.8.9
NCSA-EXT-033	Memory thresholds should be configured Memory reservation keeps the console reachable during resource exhaustion.	platform.memory_thresholds equals True	Low extended	NIST 800-53: SC-5

routing — 2 controls

Routing protocol authentication and filtering

Control	Requirement	Field & test	Severity	Framework references
NCSA-EXT-037	Routing protocol authentication must be enabled Unauthenticated routing lets an adjacent device redirect traffic.	<code>routing.ospf_auth</code> <code>equals True</code>	High extended	NIST 800-53: SC-8, IA-3 ISO 27001: A.5.10, A.5.14, A.5.33, A.8.20, A.8.26
NCSA-EXT-038	Route filtering should be applied Prefix and AS-path filters bound what a peer can inject.	<code>routing.prefix_filtering</code> <code>is set</code>	Medium extended	NIST 800-53: AC-4 ISO 27001: A.5.14, A.8.22, A.8.23

security — 2 controls

Inspection services -- IPS, antivirus, content filtering

Control	Requirement	Field & test	Severity	Framework references
NCSA-EXT-039	Intrusion prevention should be enabled IPS is the appliance's primary detection capability; disabled, the licence is unused. Remediation held for: <code>sonicwall_sonicos</code>	<code>security.ips.enabled</code> <code>equals True</code>	High extended	NIST 800-53: SI-4 ISO 27001: A.8.16
NCSA-EXT-040	Gateway anti-virus should be enabled Gateway scanning catches what endpoint controls miss in transit. Remediation held for: <code>sonicwall_sonicos</code>	<code>security.gav.enabled</code> <code>equals True</code>	Medium extended	NIST 800-53: SI-3 ISO 27001: A.8.7

services — 1 control

Services the device runs on its own behalf

Control	Requirement	Field & test	Severity	Framework references
NCSA-EXT-031	Unnecessary services must be disabled Every enabled service the device does not need is attack surface it need not have.	<code>services.unused_enabled</code> <code>is set</code>	Medium extended	NIST 800-53: CM-7 ISO 27001: A.8.19

snmp — 6 controls

SNMP version, credentials and access restriction

Control	Requirement	Field & test	Severity	Framework references
NCSA-SNMP-002	Default SNMP community strings must not be used Default community strings are the first thing an attacker tries and grant device-wide read or write access. Remediation held for: <code>cisco_iosxe_router</code> , <code>juniper_srx</code>	<code>snmp.communities</code> <code>contains_none</code> <code>public, private,</code> <code>cisco, admin,</code> <code>default</code>	Critical core	NIST 800-53: IA-5, AC-17(2) ISO 27001: A.5.14, A.5.16, A.5.17, A.6.7
NCSA-EXT-023	SNMPv3 authentication must be enabled SNMP without authentication exposes device state to anyone who can reach it. Conditional: SNMP is disabled on this device, so SNMPv3 settings are moot	<code>snmp.v3_auth</code> <code>equals True</code>	High extended	NIST 800-53: IA-3, SC-8 ISO 27001: A.5.10, A.5.14, A.5.33, A.8.20, A.8.26
NCSA-EXT-024	SNMPv3 privacy (encryption) must be enabled Authenticated but unencrypted SNMP still discloses topology and configuration. Conditional: SNMP is disabled on this device, so SNMPv3 settings are moot	<code>snmp.v3_priv</code> <code>equals True</code>	High extended	NIST 800-53: SC-8.1 ISO 27001: A.5.10, A.5.14, A.5.33, A.8.20, A.8.26

NCSA-EXT-025	SNMP access must be restricted by ACL A community string reachable from anywhere is a credential with no perimeter. Remediation held for: sonicwall_sonicos	snmp.community_acl equals True	High extended	NIST 800-53: AC-3, SC-7 ISO 27001: A.5.14, A.5.15, A.5.33, A.8.16, A.8.18, A.8.20, A.8.22, A.8.23, A.8.26, A.8.3, A.8.4
NCSA-SNMP-001	SNMP v1/v2c must not be used SNMP v1 and v2c authenticate with a cleartext community string and provide no confidentiality. Remediation held for: cisco_iosxe_router, juniper_srx, sonicwall_sonicos	snmp.version not_in 1, v1, 2c, v2c, 2	High core	NIST 800-53: AC-17(2), IA-3, SC-8 ISO 27001: A.5.10, A.5.14, A.5.33, A.6.7, A.8.20, A.8.26 STIG: V-215841, V-215842, V- 220552
NCSA-EXT-026	SNMP views should restrict the exposed MIB A restricted view limits what a compromised community string can read.	snmp.views is set	Low extended	NIST 800-53: AC-6 ISO 27001: A.5.15, A.8.18, A.8.2

time — 4 controls

Clock synchronisation and its authentication

Control	Requirement	Field & test	Severity	Framework references
NCSA-NTP-001	Time must be synchronised to authoritative sources Without synchronised time, log correlation across devices is unreliable and forensic timelines cannot be defended. Remediation held for: cisco_iosxe_router, juniper_srx	time.ntp_enabled equals True	Medium core	NIST 800-53: AU-8, AU-8(1) ISO 27001: A.8.17 STIG: V-215838, V-220549
NCSA-NTP-002	NTP must be authenticated Unauthenticated NTP allows an attacker to shift device time and invalidate or misdirect audit records. Remediation held for: cisco_iosxe_router, juniper_srx	time.authenticated equals True	Medium core	NIST 800-53: AU-8(1), SC-45 ISO 27001: A.8.17 STIG: V-215843
NCSA-EXT-027	At least two time sources should be configured A single time source is a single point of failure for every timestamp.	time.servers min_count 2	Low extended	NIST 800-53: AU-8 ISO 27001: A.8.17
NCSA-EXT-028	The device timezone must be set explicitly An unset timezone makes cross-device log correlation ambiguous.	time.timezone is set	Low extended	NIST 800-53: AU-8 ISO 27001: A.8.17

5. How a control is decided

Each control resolves to exactly one of seven states. The distinction between *not applicable* and *unknown* is the one that matters most: the first is a property of the device, the second is a limit of the assessment.

State	Meaning
PASS	Checked; the device is configured correctly.
FAIL	Checked; it is not.
PARTIAL	Some scoped instances comply and others do not.
NOT_APPLICABLE	Does not apply to this device. Excluded from scoring, and each exclusion carries a written justification.
UNKNOWN	Could not be determined from the configuration supplied. Neither a pass nor a failure, and counted in coverage.
MANUAL_REVIEW	Needs a human decision.
ERROR	The check itself failed — a defect in the tool.

The compliance score is calculated over decided controls only and is always reported beside coverage. A high score on low coverage means little of the device could be assessed, not that the device is well configured.